MIL-B-16029F
19 April 1974

SUPERSEDING
MIL-B-16029E
30 November 1970

MILITARY SPECIFICATION

BUCKETS, DRAGLINE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers dragline buckets for earth and rock excavation.

1.2 Classification. Buckets shall be of the following types, classes, and sizes as specified (see 6.2.1):

Type I - Light duty.
Type II - Medium duty.
Type III - Heavy duty.

Class P - Perforated plate.
Class S - Solid plate.

Size 37 - 3/8 cubic yard (cu yd).
Size 50 - 1/2 cu yd.
Size 75 - 3/4 cu yd.
Size 100 - 1 cu yd.
Size 150 - 1-1/2 cu yd.
Size 200 - 2 cu yd.
Size 250 - 2-1/2 cu yd.
Size 300 - 3 cu yd.
Size 350 - 3-1/2 cu yd.

2. APPLICABLE DOCUMENTS

* 2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-S-681 - Steel Castings.
RR-W-410 - Wire Rope and Strand.

FSC 3815

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MIL-T-704 - Treatment and Painting of Material.
MIL-C-3580 - Cranes and Crane-Shovels, Truck, Crawler, and Wheel Mounted, Full-Revolving, and Their Attachments, Packaging of.
MIL-S-17249 - Steel Castings, Hadfield Manganese (Low Magnetic Permeability).
MIL-T-52176 - Teeth, Surface, Ripping; (Points, Shanks, Keepers and Keeper Locks), Bucket and Dipper, Crane-Shovel.

STANDARD

MILITARY

MIL-STD-129 - Marking for Shipment and Storage.

(Copies of specifications and standard required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

* 2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY OF TESTING AND MATERIALS (ASTM)

A36 - Specification for Structural Steel.
A391 - Specification for Alloy Steel Chain.
A514 - Specification for Alloy Steel Plate.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

NATIONAL MOTOR FREIGHT TRAFFICE ASSOCIATION, INC., AGENT

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Association, Inc., Tariff Order Section, 1616 P Street, N. W., Washington, DC 20036.)

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UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Tariff Publishing Officer, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal Agencies.

3. REQUIREMENTS

3.1 Description. The bucket unit shall include the bucket body, hoist and drag chains, spreader bar, dump block, dump rope, teeth, sheaves, and all necessary sockets, links, shackles, rings, and pins. The bucket unit shall be delivered complete and ready for use.

* 3.2 First article. When specified (see 6.2.1), the contractor shall furnish a bucket for first article inspection and approval (see 4.2.1 and 6.3).

* 3.3 Standard bucket components and accessories. Except as modified by this specification, the bucket components, and accessories shall be the manufacturer's standard or optional items, which meet or exceed the requirements of this specification.

* 3.4 Material. All material shall be new and unused. All material shall be as specified herein. Material not specified shall be of the same quality used for the intended purpose in commercial practice. Materials used shall be free from defects which would adversely affect the performance or maintainability of individual components or the overall assembly.

* 3.5 Capacity and weight. The struck level capacity of the bucket shall be within 2 percent of that specified in table I, and shall be established by taking the actual side area, with the forward line delineated by the angle of repose of the material assumed at 90° from the deepest part of the cutting edge, multiplied by the average inside bucket width, which is determined by measurements of the top and bottom of both the front and rear reference points of the bucket. The weight of class S buckets, as delivered, shall be not less than specified for the respective size in table I. In addition, the delivered weight of class S buckets of size 37 through size 100 shall not exceed the respective specified weight by more than 10 percent, and the delivered weight of class S buckets

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of size 150 through size 350 shall not exceed the respective specified weight by more than 12 percent. The delivered weight of class P buckets shall be not less than 97 percent of the weight specified in table I, for the respective size of class S buckets. The bucket body, including teeth, hitch, plate pines, trunnion pins, and anchor pins shall be not less than 70 percent of the total weight of the bucket.

Table I. Capacities and weights (weights based on class S buckets)

	Size								
	37	50	75	100	150	200	250	300	350
Struck capacity (cu ft)	11.25	15	22.5	30	45	60	75	90	105
Type I, weight (lb)	875	1,000	1,420	1,720	2,550	3,200	3,620	4,120	4,650
Type II, weight (lb)	990	1,150	1,550	1,830	2,930	3,550	4,100	4,800	5,350
Type III, weight (lb)	1,120	1,350	1,850	2,260	3,220	4,100	4,850	5,500	6,150

3.6 Components.

* 3.6.1 Bucket body. The bucket body shall include bottom, side, and back plates, with partial or complete lip and bottom, dependent on manufacturer's commercial practice. Method of connecting bucket body assembly of arch, lip or complete bottom, hoist trunnions, hitch plates, runner bars, heel plates, and corner wearing shoes shall be either cast integral or welded, or combination of both. The body shall be designed in accordance with best commercial practice to handle the class of excavation for which it is intended. The inside surfaces shall be smooth and free from projections. The connections between the bottom and side plates shall have a radius and the back plates shall be curved sufficiently to provide satisfactory loading and dumping performance. Cast lips or cast bottoms and connections to body and side plates, shall provide a firm joint that is consistent with industry practice for weldments or riveted components. Inside clear width at the lip shall be slightly larger than at the heel to facilitate the discharge of adhesive material. The plates for the bottom, sides, and back of the bucket shall be of steel which meets or exceeds the requirements of ASTM designation A36. The back and side plates of class P buckets shall be perforated in a manner that will not impair or weaken the bucket strength.

* 3.6.2 Arch. The front end components shall be of cast or rolled plate steel construction. The arch shall be designed so that it will withstand its proportionate share of the load during all operation cycles without deformation or failure, provided it is not subjected to loading or digging

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forces in excess of four times the breaking strength of the size dump rope specified in table II. Ample clearance shall be provided for the flow of excavated material in and out of the bucket.

Table II. Dump rope size (inches)

	Bucket size								
	37	50	75	100	150	200	250	300	350
Rope size	1/2	5/8	5/8	5/8	3/4	3/4	7/8	1	1-1/8

* 3.6.3 Lip. The lip shall be cast manganese steel conforming to MIL-S-17249, cast steel conforming to QQ-S-681 (class 105-85), or alloy steel plate conforming to ASTM Designation A514, or better. The lip shall form the cutting edge. The extent of the lip shall conform to the manufacturer's standard practice, and shall accommodate the teeth as specified (see 3.6.4). All lips or cutting edges shall enable replacement of expendable components.

* 3.6.4 Teeth. Unless otherwise specified (see 6.2.1), manufacturer's standard bucket teeth shall be furnished. When specified (see 6.2.1), teeth shall conform to MIL-T-52176. The teeth shall be readily replaceable by the use of standard tools.

3.6.5 Hitch plates. The hitch plates for attaching the drag chains to the bucket shall be of one of the steels specified in 3.6.3 except for type I and II buckets of size 100 and smaller, the steel may conform to QQ-S-681, class 65-35. The hitch plates shall provide not less than two positions for the attachment of the drag chains as an aid in controlling the digging depth. The plates shall be either riveted or welded to the bucket body.

* 3.6.6 Trunnion brackets. The trunnion brackets shall be of one of the steels specified in 3.6.3. The trunnion brackets shall be accurately located on the bucket body to provide proper balance, and shall be either riveted or welded in place.

* 3.6.7 Runners. Renewable runners shall be provided and shall be either riveted or welded to the heel of the bucket. The runners shall be of one of the steels specified in 3.6.3.

* 3.6.8 Heel plates. Renewable or integral cast type heel plates shall be provided. Renewable heel plates shall be either riveted or welded between the runner bars at the heel of the bucket. The plates shall be of one of the steels specified in 3.6.3.

* 3.6.9 Corner wearing shoes. All buckets shall have renewable cast corner wearing shoes of the material specified in 3.6.3. Renewable wearing shoes shall be bolted, riveted, or welded to the bottom corner.

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* 3.6.10 Chains. Unless otherise specified (see 6.2.1), all buckets shall be furnished with the manufacturer's standard hoist and drag chains, links, rings, and shackles. When specified (see 6.2.1), chains shall meet or exceed the requirements of ASTM Designation A391.

* 3.6.11 Spreader bar. Spreader bar shall be of rolled or cast steel with a minimum safety factor of 3, based on yield strength, using a static struck bucket loaded with material weighing 100 pounds per cubic foot.

3.6.12 Dump block. The dump block assembly shall include the frame, sheave, axle, and lubrication fittings. The frame shall be of cast or rolled steel. The sheave shall be made of steel conforming to one of the steels specified in 3.6.3 and shall have smooth grooves with a pitch diameter of not less than 16 times the diameter of the dump rope. The sheave shall be furnished with lubricated roller type bearings, or needle bearings as normally provided on commercial buckets of the size indicated. The sheave pin shall be of cold rolled or forged steel.

3.6.13 Dump rope. The dump rope shall be 6 by 19 improved plow steel, uncoated, preformed, independent wire rope core conforming to RR-W-410, type I, class 2, construction 3. Size of dump rope shall be not less than specified in table II (see 3.6.2). Sockets shall be of steel conforming to one of the steels specified in 3.6.3.

3.7 Cleaning, treatment, and painting. Unless otherwise specified (see 6.2.1), surfaces normally painted in good commercial practice shall be cleaned, treated, and painted as specified herein. When specified (see 6.2.1), the unit shall be cleaned, treated, and painted in accordance with MIL-T-704, type A. In either case, color of the finish coat shall be as specified (see 6.2.1).

3.7.1 Cleaning. Metal surfaces to be painted shall be cleaned to insure that they are free from all oil, grease, welding slag and spatter, all mill scale, products of corrosion, dirt, or other foreign substances.

3.7.2 Treatment. As soon as practicable after cleaning, and before any evidence of rust or other contamination can result, metal surfaces shall be treated with a primer pretreatment coating to increase the adhesion of the coating system.

3.7.3 Painting. Painting shall consist of at least one coat of rust-inhibiting primer and one coat of finish enamel. The rust-inhibiting primer shall be applied to a clean, dry surface as soon as practicable after cleaning and treating. Painting shall be with manufacturer's current materials according to manufacturer's current processes except that the total dry film thickness shall be not less than 2.5 mils. The paint shall be free from runs, sags, orange peel, or other defects.

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* 3.8 Nameplate. A nameplate shall be furnished by the contracting officer for each unit. The contractor shall stamp all necessary data in the blank spaces provided for that purpose and securely affix it to each unit in a conspicuous place with rivets or nonferrous screws or bolts not less than 1/8 inch in diameter. The applicable nomenclature contained in contract item description shall be placed in the top blank.

* 3.9 Lubrication. Lubrication fittings shall be provided in accessible protected positions for bearings and moving parts requiring lubrication.

3.10 Workmanship. All parts, components, and assemblies of the bucket, including castings, forgings, molded parts, stamping, bearings, seals, machined surfaces, and welded parts, shall be clean and free from sand, dirt, fins, pits, sprues, scale, flux, and other harmful extraneous material. External surfaces shall be free from burrs, sharp edges, and corners except when sharp edges and corners are required.

* 3.10.1 Steel fabrications. The steel used in fabrication shall be free from kinks, sharp bends, and other conditions which would be deleterious to the finished product. Manufacturing processes shall not reduce the strength of the steel to a value less than intended by the design. Manufacturing processes shall be done neatly and accurately. All bends shall be made by controlled means to insure uniformity of size and shape.

* 3.10.2 Bolted connections. All bolt holes shall be accurately punched or drilled and shall have the burrs removed. Washers or lockwashers shall be provided. All nuts, bolts, and screws shall be tight.

* 3.10.3 Riveted connections. All rivet holes shall be accurately punched or drilled and shall have the burrs removed. Rivets shall completely fill the hole. Rivet heads shall be of an approved shape and shall be concentric with the rivet. Rivet head shall make full contact with the intended surface. The rivets shall be so fabricated as to develop a joint strength not less than the design value.

* 3.10.4 Welders and welding. The surface of parts to be welded shall be free from rust, scale, paint, grease, or other foreign matter. Spot, tack, or intermittent welds for strength will not be permitted. Weld penetration shall be such as to provide transference of maximum design stress through the base metal juncture. Fillet welds shall be provided when necessary to reduce stress concentration. Unless otherwise specified (see 6.2.1), welding procedures shall be in accordance with a nationally recognized welding code. When specified (see 6.2.1), before assigning any welder to manual welding work, the manufacturer shall certify to the contracting officer that the welder has passed the qualification tests as prescribed by either the American Welding Society or the American Society of Mechanical Engineers for the type of welding operation to be performed.

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* 3.10.5 Castings. All castings shall be sound and free from patching, misplaced coring, warping, or any other defect which reduces the castings ability to perform its intended function.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Examination. Each bucket shall be examined for compliance with the requirements specified in section 3 of this specification. Any redesign or modification of the supplier's standard product to comply with specified requirements, or any necessary redesign or modification following failure to meet specified requirements, shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all visual examinations and dimensional measurements. Noncompliance with any specified requirements or presence of one or more defects preventing or lessening maximum efficiency shall constitute cause for rejection.

* 4.2.1 First article inspection. First article inspection shall be performed on one bucket when a first article sample is required (see 3.2). This inspection shall include the examination of 4.2 and the tests of 4.3. The first article may be a standard production item from the supplier's current inventory provided the bucket meets the requirements of the specification and is representative of the design, construction, and manufacturing technique applicable to the remaining buckets to be furnished under the contract.

4.3 First article tests. When first article is required, the first article bucket of each type, class, and size shall be subjected to the tests specified in 4.3.1 and 4.3.2. The first article tests shall be performed by the supplier. The supplier shall provide all material and equipment to perform all testing specified. Failure to pass any phase of the required tests shall be cause for the Government to refuse acceptance of all buckets until corrective action has been taken.

4.3.1 Material rehandling test. Bucket shall be tested for rehandling of materials. Materials for testing shall be loose or stockpiled earth, rock, or a combination thereof. The bucket shall scoop up a load, raise,

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and dump contents completely. This test cycle shall be repeated not less than 10 times and the average load shall be not less than the struck capacity required in 3.5.

4.3.2 Digging test. Bucket shall be subjected to a digging test. Materials for testing shall be in-place moist earth, rock, or combination of earth and rock. Bucket shall be set down or dropped on the materials at the proper angle to dig and scoop a load immediately. The bucket shall be raised and conveyed to the proper location where material shall be completely dumped. This test cycle shall be repeated not less than 10 times and the average load shall be not less than the struck capacity of the bucket as required in 3.5. The loading and dumping of any materials being tested shall not be retarded by protrusions or projections inside the bucket.

4.4 Production unit tests. The contractor's testing procedures shall, as a minimum, assure that each bucket is capable of meeting the requirements of this specification.

4.5 Preparation for delivery inspection. The preservation, packaging, packing, and marking of the buckets shall be inspected to verify conformance to the requirements of section 5.

5. PREPARATION FOR DELIVERY .

5.1 Preservation and packaging. The preservation and packaging shall be level A or C, as specified (see 6.2.1).

5.1.1 Level A. The complete buckets shall be preserved and packaged in accordance with level A of MIL-C-3580.

5.1.2 Level C. The complete buckets shall be preserved and packaged in a manner that will insure adequate protection against deterioration and damage during shipment. This level may conform to the supplier's commercial practice when such meets the requirements of this level.

5.2 Packing. The packing shall be level A, B, or C, as specified (see 6.2.1).

5.2.1 Level A. The complete buckets shall be packed in accordance with level A of MIL-C-3580.

5.2.2 Level B. The complete buckets shall be packed in accordance with level B of MIL-C-3580.

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5.2.3 Level C. The complete buckets shall be packed in a manner which will insure arrival at destination in satisfactory condition and which will be acceptable to the carrier at lowest rates. Packing shall comply with Uniform Freight Classification rules or National Motor Freight Classification rules.

5.3 Marking. The buckets, interior packages, and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. Dragline buckets covered by this specification are intended for excavating earth and loose rock.

6.1.1 Type I. The type I, light duty bucket is intended for excavating sand, loose earth, loam, moist and wet clay, and loose small gravel.

6.1.2 Type II. The type II, medium duty bucket is intended for excavating dry clay, loose boulders, coal, gravel, and soft shale.

6.1.3 Type III. The type III, heavy duty bucket is intended for excavating average cemented gravel, hard shale, conglomerate hardpan, caliche, and broken rock.

6.1.4 Class P. The class P perforated bucket is intended for excavating wet material that is not too fluid. This class bucket has less weight than the class S solid plate bucket.

6.2 Ordering data. Procurement documents should specify:

* 6.2.1 Procurement requirements.

(a) Title, number, and date of this specification.
(b) Type, class, and size of bucket required (see 1.2).
(c) When first article is required for inspection and approval (see 3.2, 4.2.1, and 6.3).
(d) When teeth shall conform to MIL-T-52176 (see 3.6.4).
(e) When chains shall meet or exceed the requirements of ASTM Designation A391 (see 3.6.10).
(f) When cleaning, treatment, and painting in accordance with MIL-T-704, type A is required (see 3.7).
(g) Finish coat color required (see 3.7).
(h) When certification of welder's qualification is required (see 3.10.4).
(i) Level of preservation and packaging and level of packing required (see 5.1 and 5.2).

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* 6.2.2 Contract data requirements. When this specification is used in a procurement which incorporates a DD Form 1423 and invokes the provisions of paragraph 7-104.9 (n) of the Armed Services Procurement Regulation, the data requirements identified below will be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the Contract Data Requirements List (DD Form 1423) incorporated into the contract. When the provisions of ASPR 7-104.9 (n) are not invoked, the data specified below shall be delivered in accordance with the contract requirements. Deliverable data required by this specification is cited in the following paragraph:

Paragraph	Data Requirements	Applicable DD 1664
3.10.4	Certificate of compliance	UDI-E-24007

(Copies of Data Item Descriptions required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

* 6.3 First article. When a first article is required, it shall be tested and approved under the appropriate provisions of paragraph 7-104.55 of the Armed Services Procurement Regulations. The first article should be a first production item consisting of one complete bucket unit or it may be a standard production item from the contractor's current inventory as specified in 4.2.1. The contracting officer should include specific instructions in all procurement instruments, regarding arrangements for examinations, test, and approval of the first article.

6.4 Identification of changes. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:

Army - ME
Navy - YD
Air Force - 84

Preparing activity:

Navy - YD

Project No. 3815-0087

Review activity:

DSA - CS

User activity:

Navy - MC

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DOCUMENT IDENTIFIER AND TITLE

MIL-B-16029F - BUCKETS, DRAGLINE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

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☐ YES ☐ NO *(If "Yes", in what way?)*

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